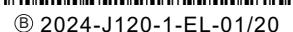


270 marks



Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

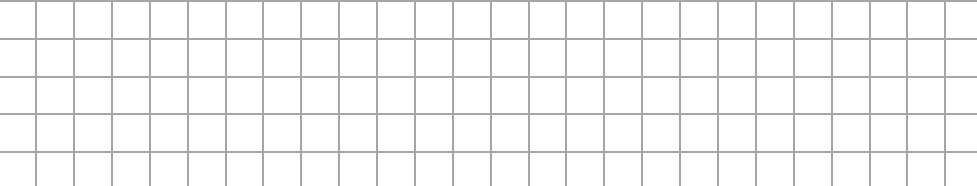
You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

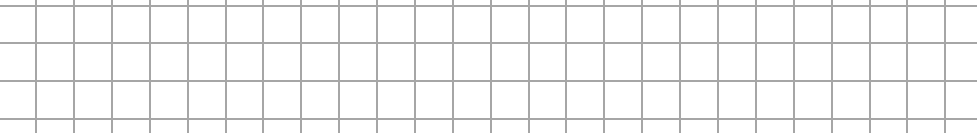
Question 1 (Suggested maximum time: 5 minutes)

- Note: \$1 billion = 1000 × \$1 million

- Express this reduced lump sum as a **percentage** of the total jackpot.

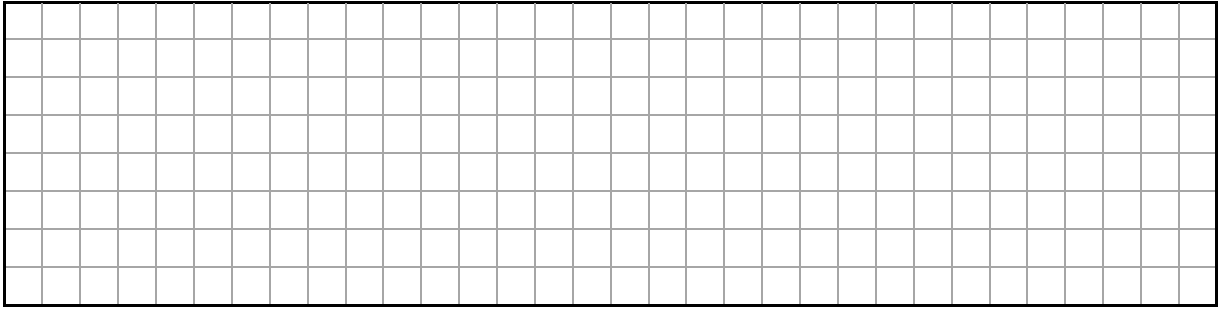


- Work out in **euro** the value of the largest national lottery jackpot.

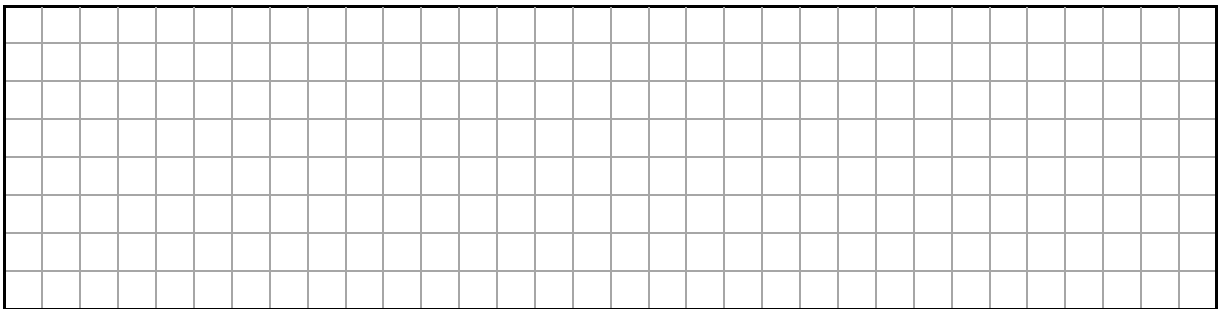


Question 2**(Suggested maximum time: 5 minutes)**

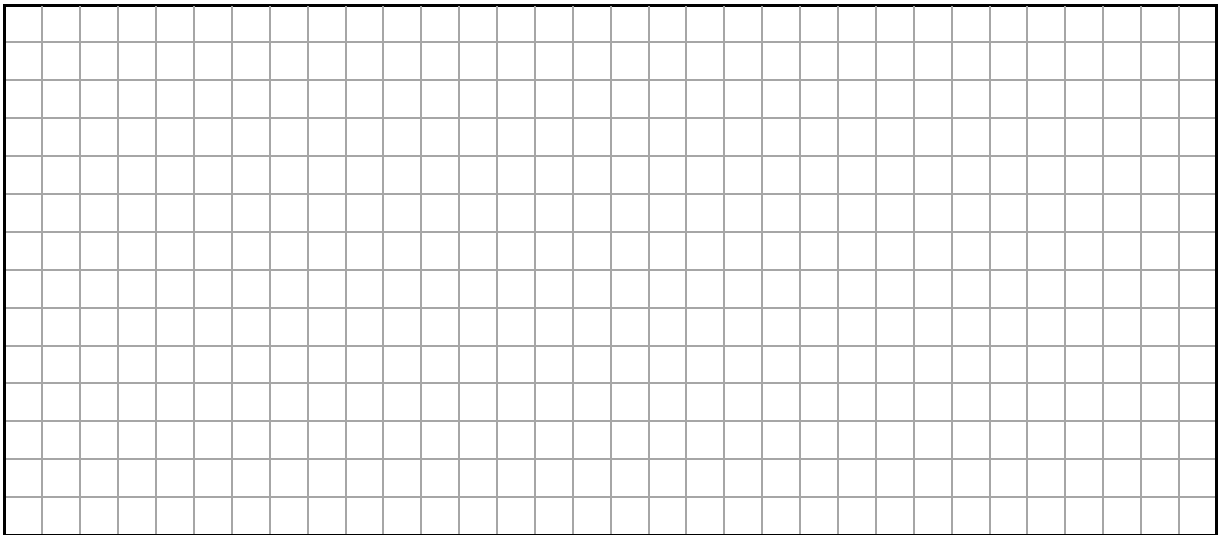
- (a) Explain why $-10 \cdot 3$ is not an integer.



- (b) Write $\frac{1}{125}$ in the form 5^k , where $k \in \mathbb{Q}$.



- (c) Write $\frac{(a^3 \times a^5)^2}{a^{10}}$ in the form a^k , where $k \in \mathbb{Q}$.



Question 3 (Suggested maximum time: 10 minutes)

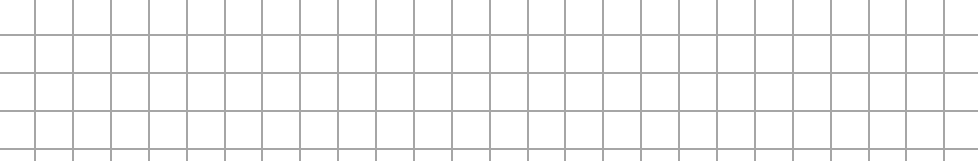
Question 3 (Suggested maximum time: 10 minutes)

- (a)** Factorise fully $3x^2 - 6x$.

A large empty grid for drawing or writing, consisting of 20 columns and 10 rows of squares.

- (b)** Write the following as a single fraction in its simplest form:

$$\frac{5x-7}{3} - \frac{x+6}{4}.$$



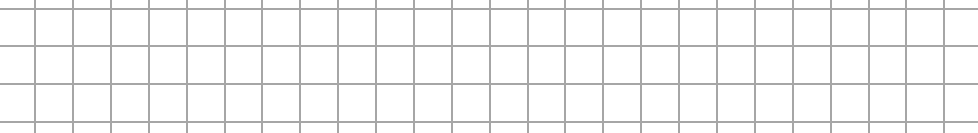
- (c) Three consecutive natural numbers sum to 102.

Let the smallest number = n , where $n \in \mathbb{N}$.

- (i)** Write an algebraic expression to represent the information above.

[illegible]

- (ii)** Solve your algebraic expression to work out the three natural numbers.

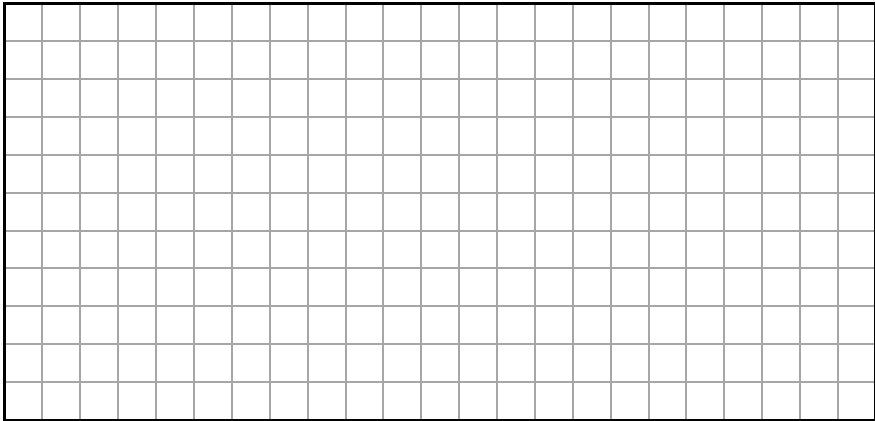


Question 4

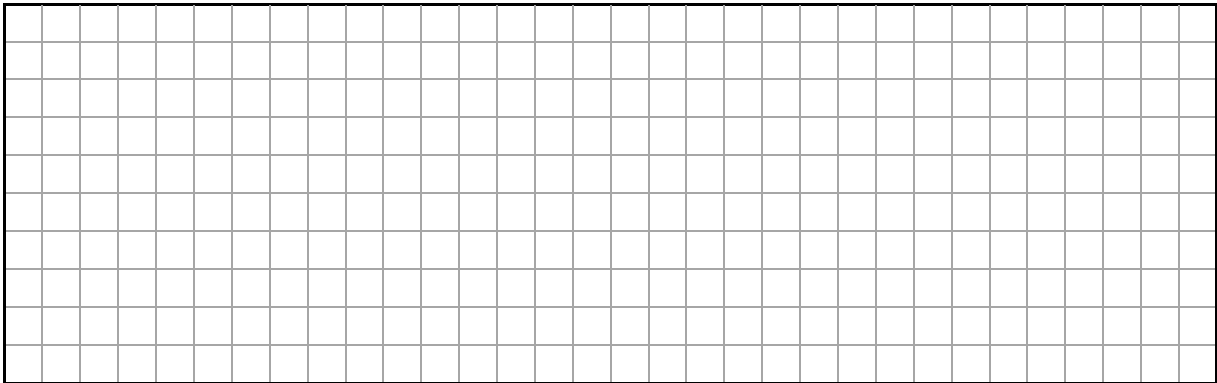
(Suggested maximum time: 15 minutes)

A rim of a basketball hoop has a diameter of 46 cm, as shown in the diagram.

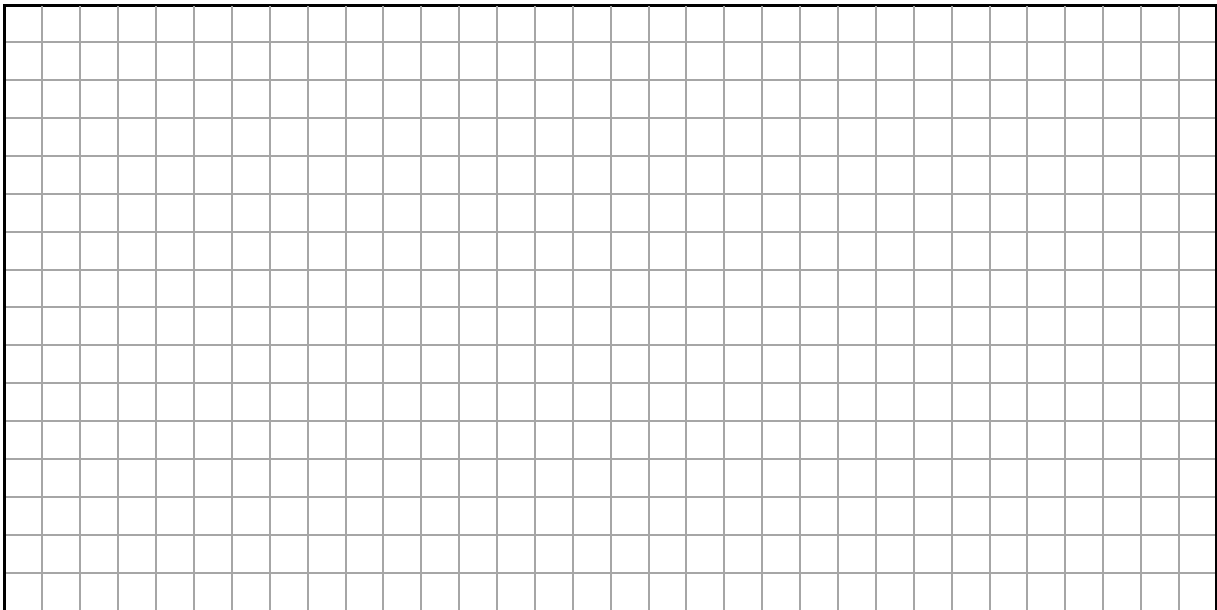
- (a) Work out the **area** enclosed by the rim.



- (b) The volume of an adult size 7 basketball is $2304\pi \text{ cm}^3$.
Work out the **radius** of the basketball.

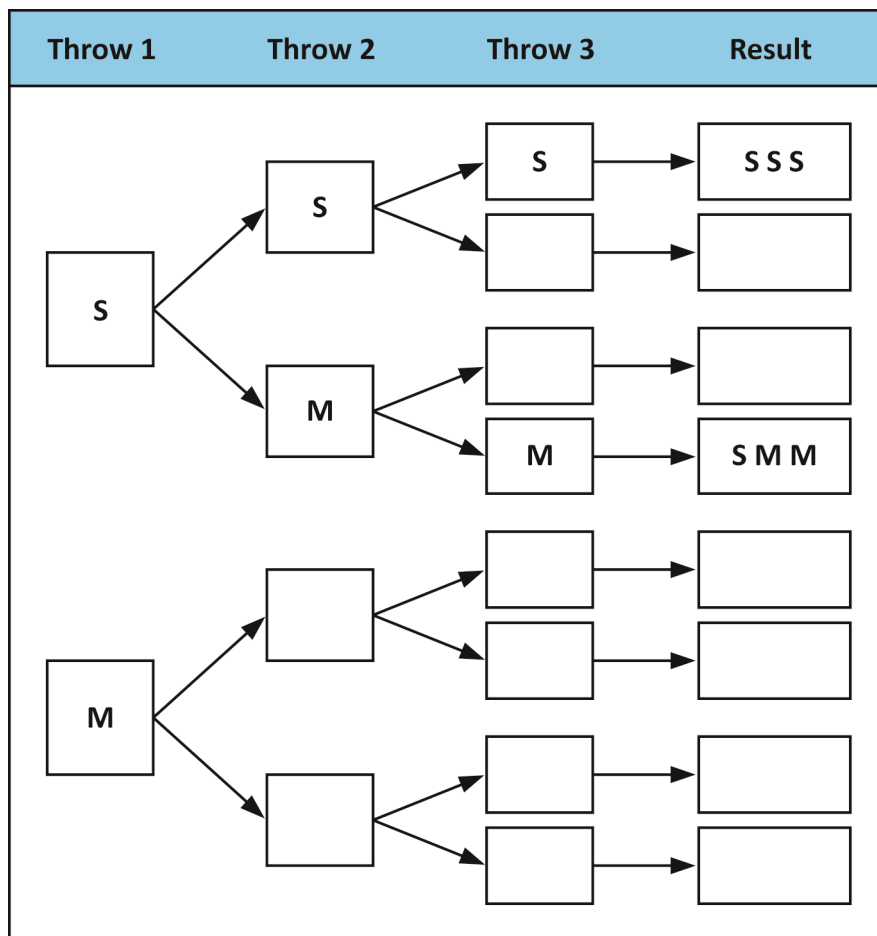


- (c) Work out the area of space inside the rim when the widest part of the basketball passes through the rim. Give your answer in cm^2 correct to one decimal place.



- (d) Peter makes three attempts to throw the basketball through the hoop. He will either score (**S**) or miss (**M**).

Complete the tree diagram below to show all possible outcomes.



Each result in the tree diagram is equally likely.

- (e) Find the **probability** that Peter **scores once only** over the three attempts.

[illegible]

- (f) Find the **probability** that Peter scores more times than he misses over the three attempts.

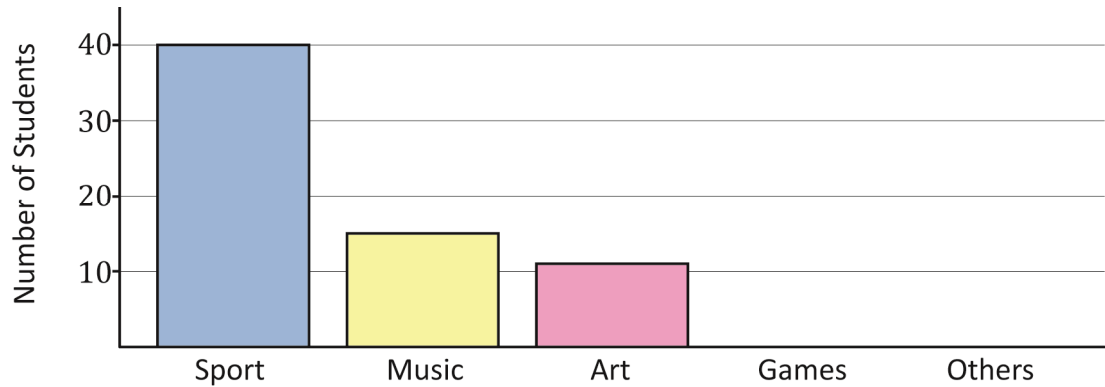
[illegible]

Question 5

(Suggested maximum time: 15 minutes)

The student council in a school carried out a survey on students’ interests.

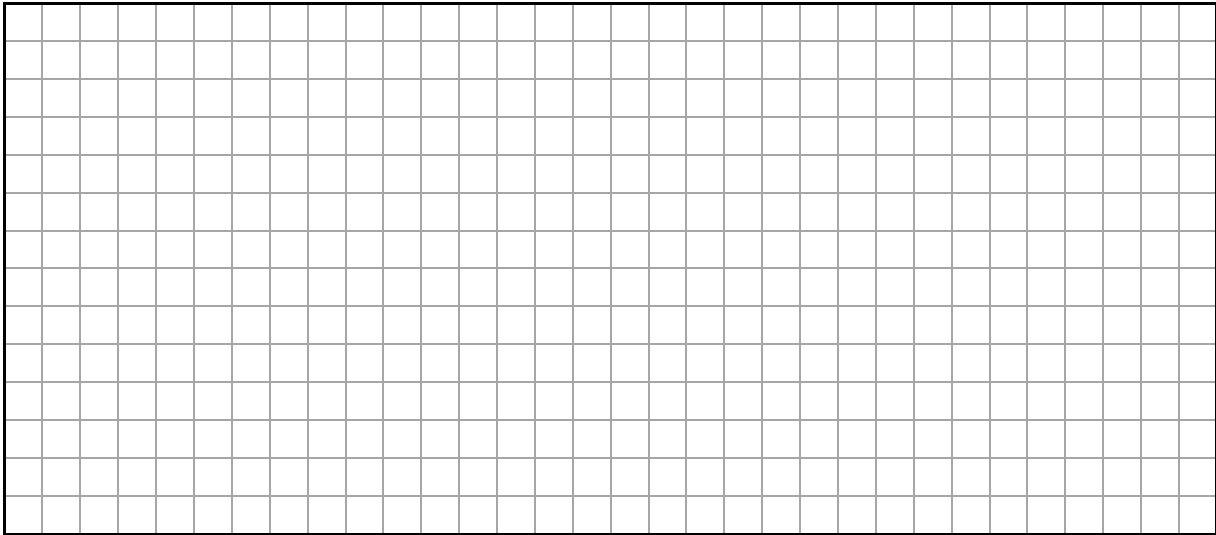
- (a) 108 students indicated their favourite school club. Some of the results are shown in the bar chart below.



- (i) Complete the table below to show the number of students who indicated their favourite club.

Club	Sport	Music	Art	Games	Others
Number of students			11	$2x$	x

- (ii) Work out the value of x in the table and use your answer to complete the bar chart above.



- (b) Another question in the survey asked the same 108 students to indicate what subject they would like to see introduced to their school.

36 students indicated Physical Education.

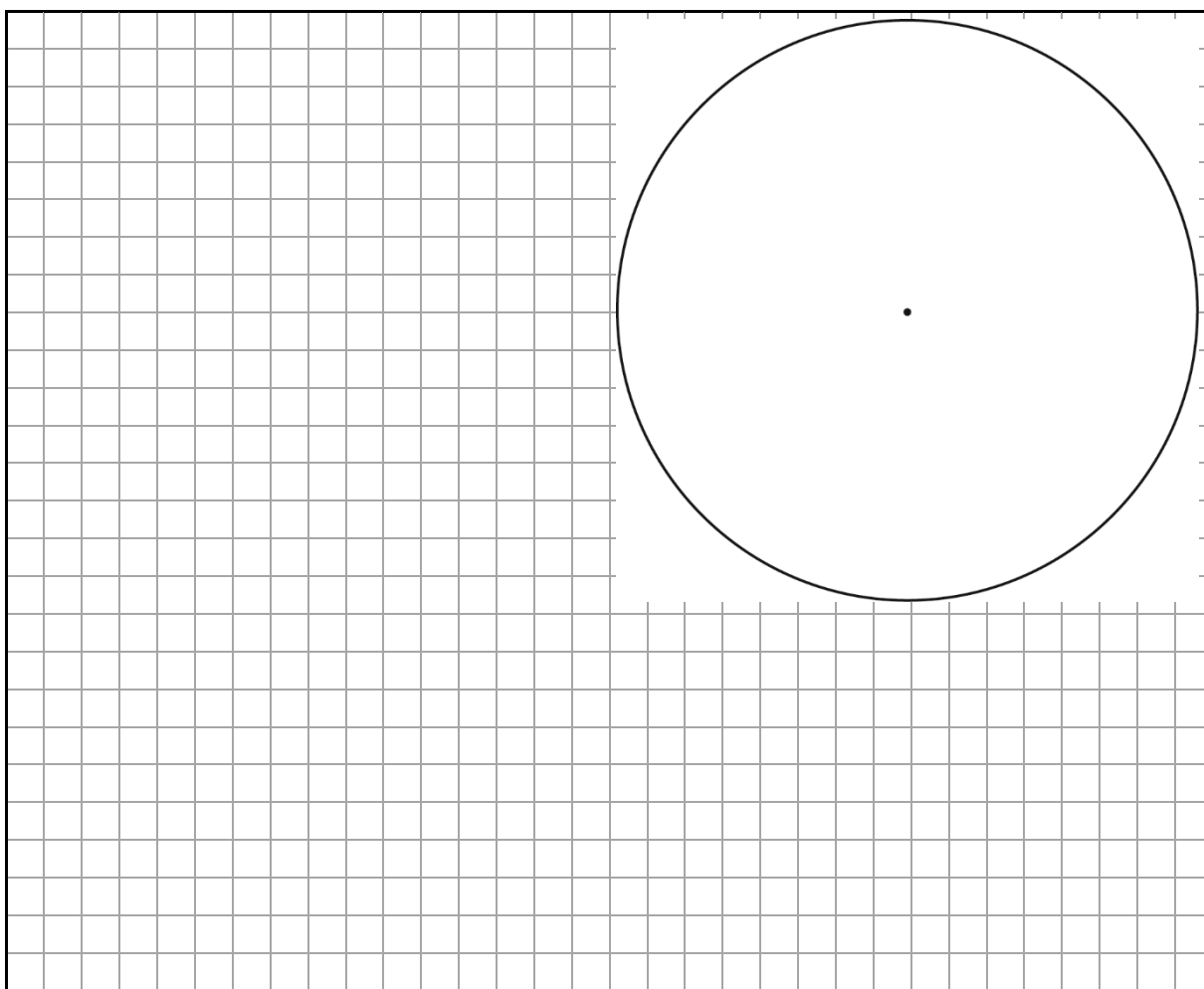
25% of students chose Politics and Society.

$\frac{1}{9}$ of the students indicated Agricultural Science.

The remaining students indicated Computer Science.

Complete the **pie chart** below to summarise the data above, showing the proportion of students that indicated Physical Education, Politics and Society, Agricultural Science and Computer Science.

Label each sector and the size of the angle clearly. Show any working out and construction lines.



This question continues on the next page

- (c) Write five natural numbers in ascending order (lowest to highest) in the boxes below so that the five numbers have:

a median of 7

a mode of 9

a range of 8

a mean of 6

Answer:

--

,

--

,

--

1

and

[illegible]

Question 6

(Suggested maximum time: 5 minutes)

A housing development has 124 houses for sale.

Some houses have the following features included:

Extra family room (F), Attic conversion (C), Extended kitchen (K).

$$\#(F \cap C \cap K) = 10.$$

$$\#(F \cap C) = 22.$$

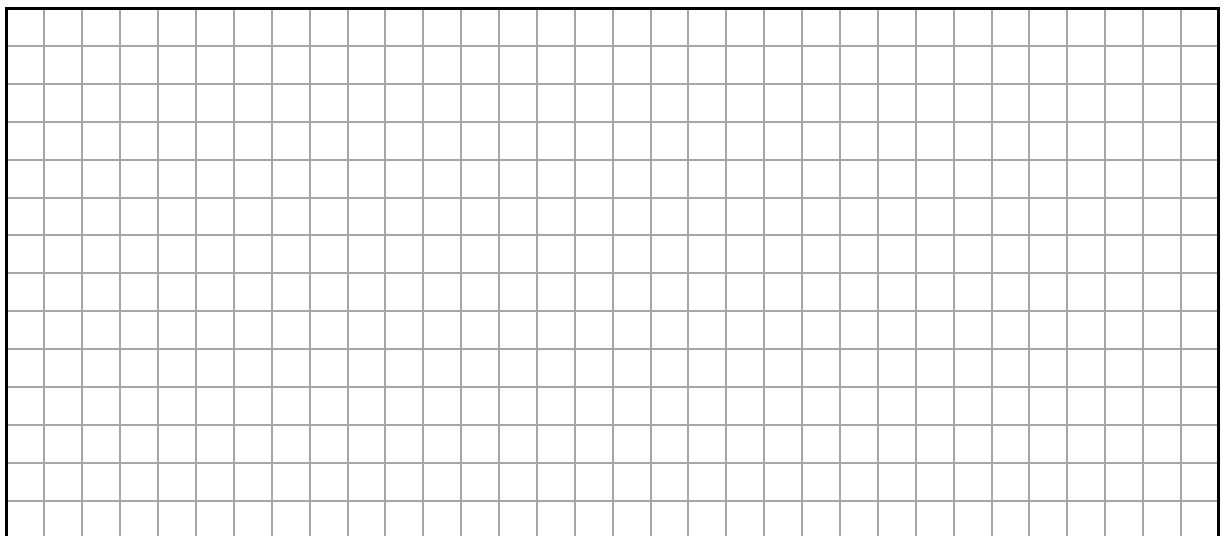
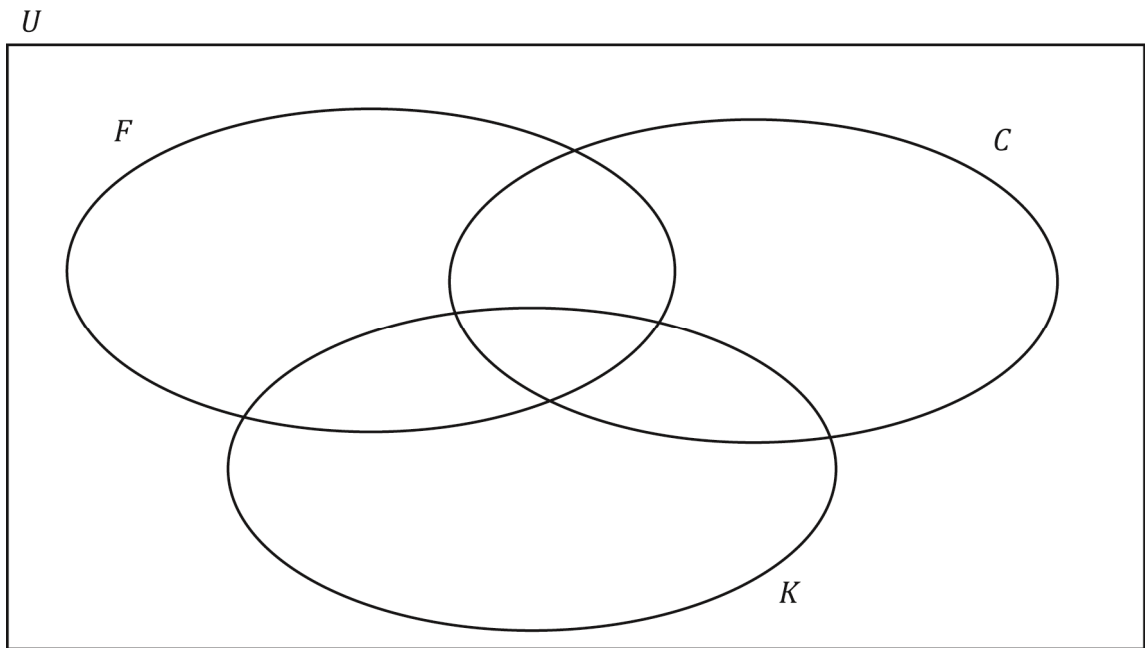
$$\#(F \cap K) = 26.$$

$$\#((C \cap K) \setminus F) = 11.$$

$$\#(F \cup C \cup K)' = 15.$$

Equal numbers of houses have only one of these features.

Using a Venn diagram, or otherwise, find the number of houses with only one feature.

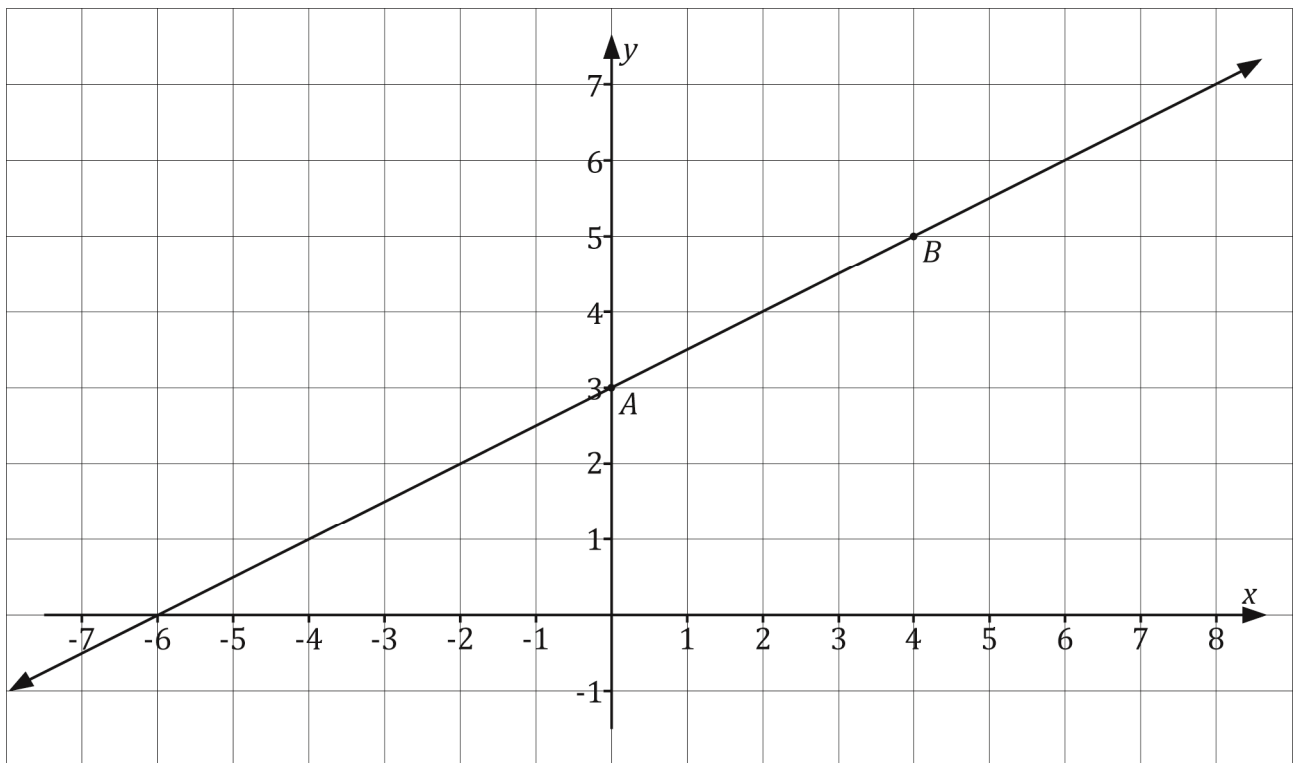


Question 7

(Suggested maximum time: 15 minutes)

The line l intersects the y -axis at the point A .

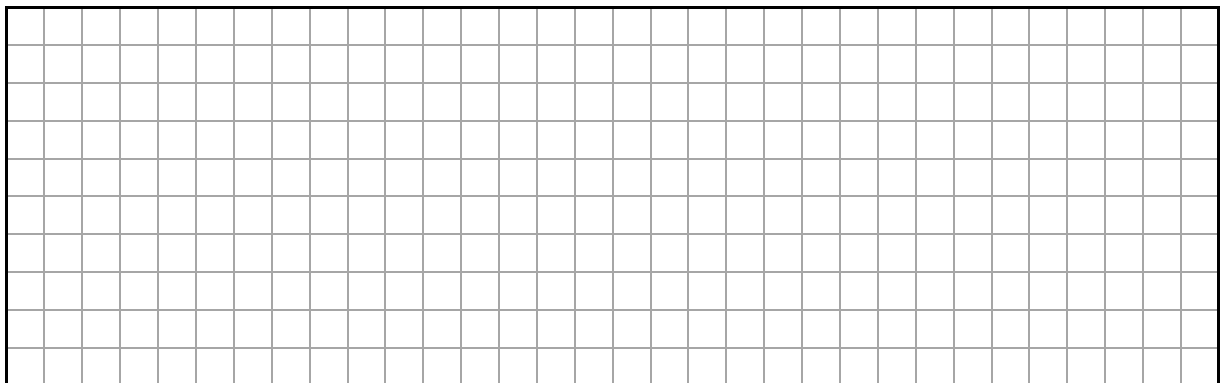
B is also a point on the line l , as indicated on the co-ordinate diagram below.



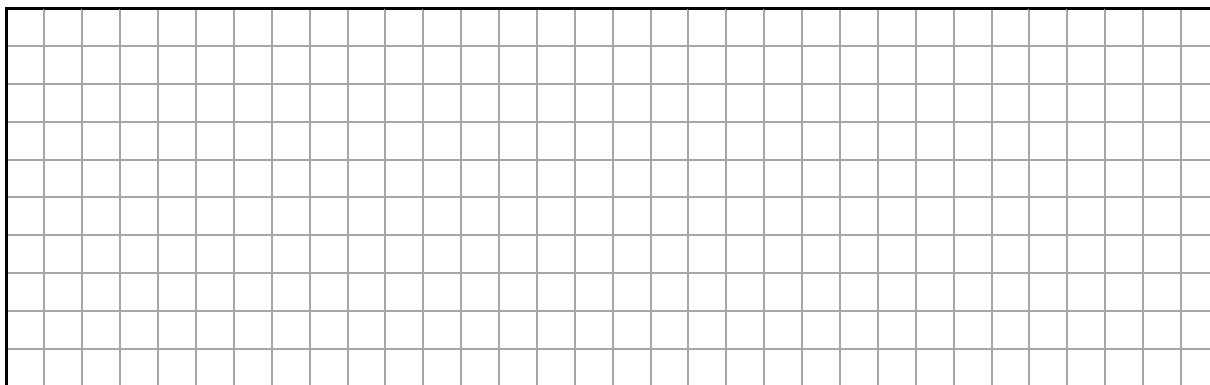
- (a) Write the co-ordinates of the points A and B in the spaces below.

$$A = \left(\quad , \quad \right) \quad B = \left(\quad , \quad \right)$$

- (b) Find the **slope** of the line l .

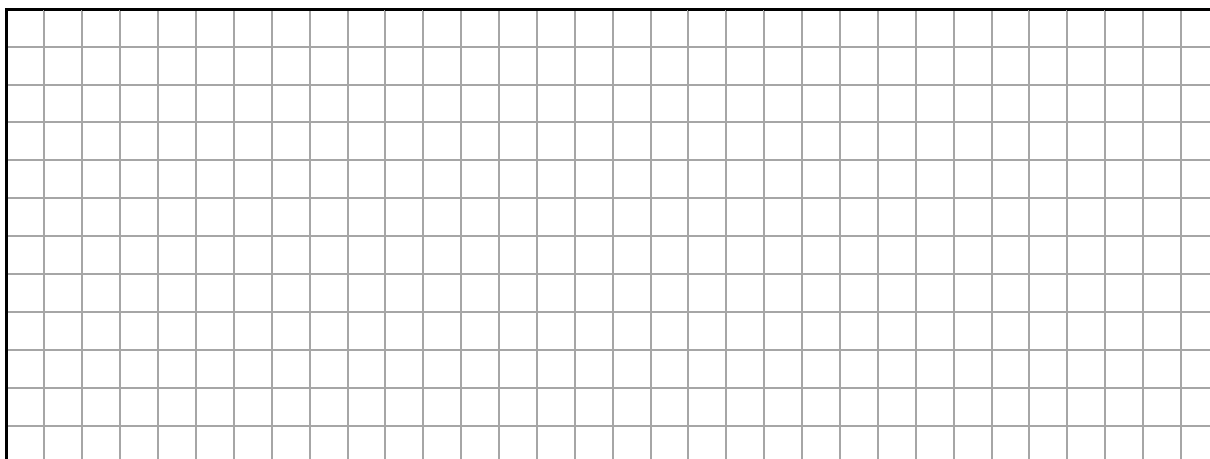


- (c) Work out the equation of the line l .
Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{R}$.



- (d) The line m is **perpendicular** to l and passes through the point $(6, 1)$.
Construct the line m on the co-ordinate diagram above.
Show all of your construction lines clearly.

- (e) Work out the equation of the line m .
Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{R}$.



- (f) Use co-ordinate geometry methods to show that $l \cap m = B$.



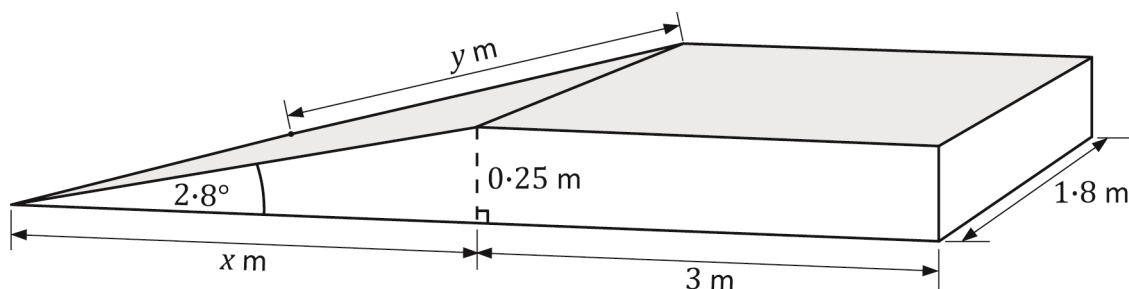
Question 8

(Suggested maximum time: 15 minutes)

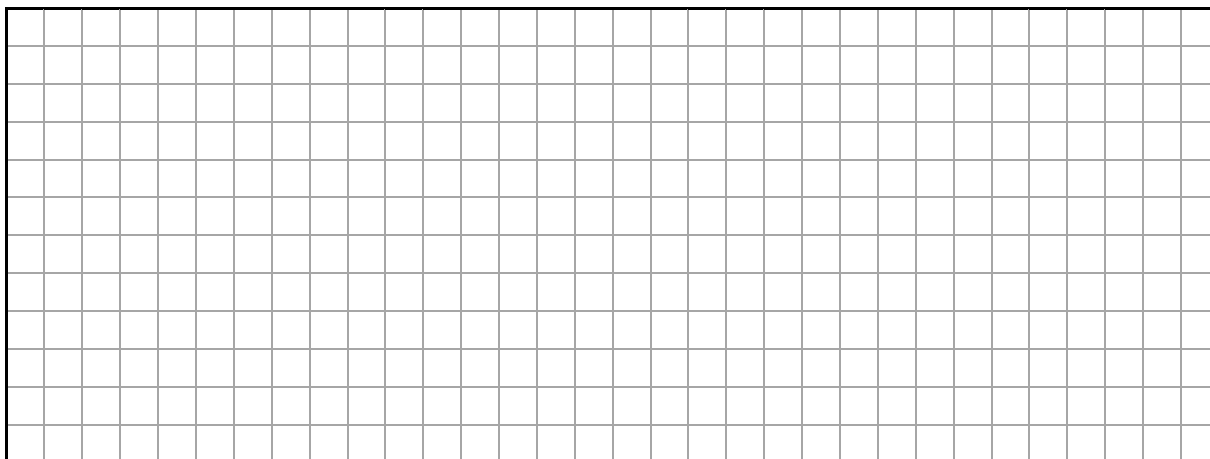
The diagram below shows a concrete ramp (diagram not to scale)

made up of a cuboid and a right-angled triangular prism.

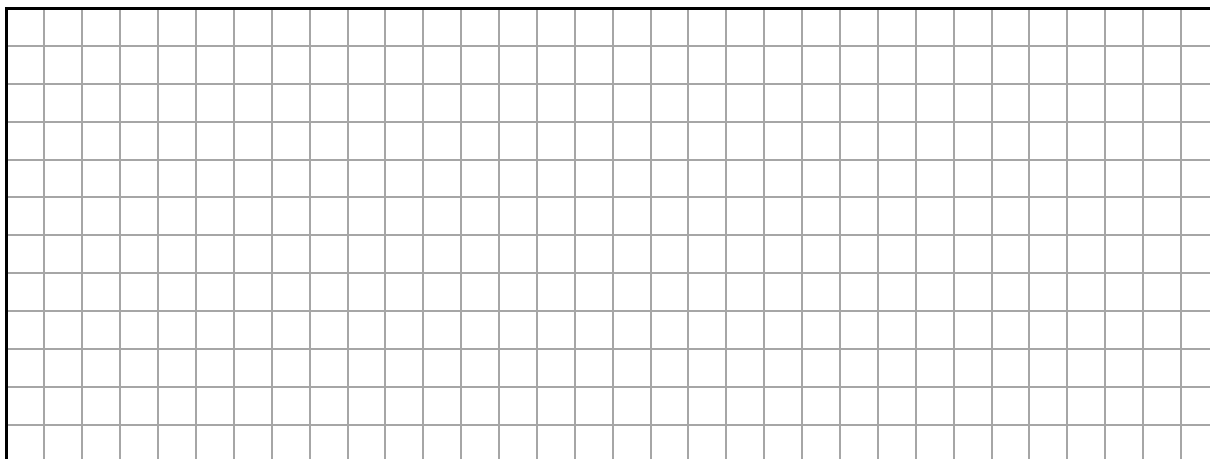
The ramp has a width of 1.8 m and a height of 0.25 m. The angle of elevation of the ramp is 2.8° .



- (a) Use trigonometry to work out the value of x on the diagram.
Give your answer correct to one decimal place.



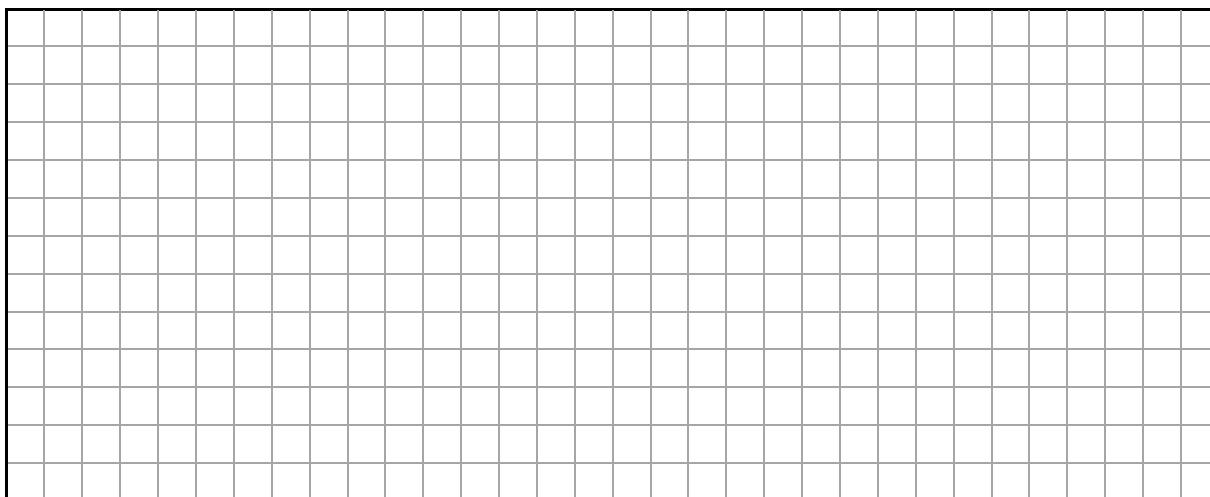
- (b) The ramp is made of concrete. Concrete is made using water, cement, sand and gravel which are mixed in the ratio 1:2:3:4 respectively.
2.5 m³ of concrete is required to make the ramp.
Work out how much water is required to make this amount of concrete.



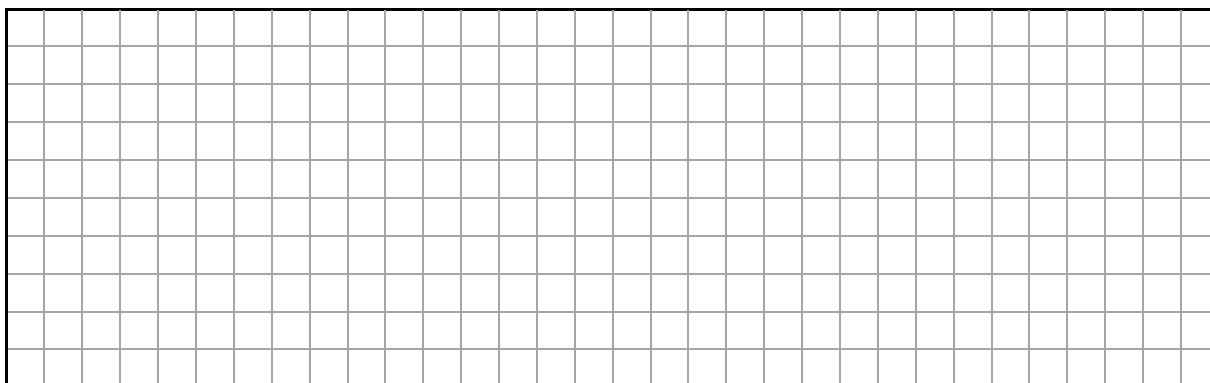
- (c) Use the **Theorem of Pythagoras** to work out the value of y on the diagram.
Give your answer correct to two decimal places.



- (d) The shaded region in the diagram shows the surface of the ramp.
This area needs a slip-resistant covering which costs €74 per m^2 .
Work out the cost of covering the surface of the ramp.



- (e) 3 workers are expected to complete the ramp in 3 days.
However, at the time of starting the job, only 2 workers are available.
How long will it take the 2 workers to complete the ramp?
Note: Assume the workers work at the same speed.



Question 9

(Suggested maximum time: 5 minutes)

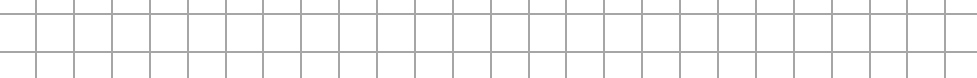
Peter says: "The initial growth of a bacterium can be modelled using an exponential function".

A group of students set up an experiment to grow a particular bacterium to verify this statement. The numbers of bacteria present after the experiment begins are shown in the table below.



Time (hours)	0	1	2	3	4
Number of bacteria	1	3		27	

- (a) Given that the number of bacteria grew as expected, fill in the two missing values in the table above.



- (b)** Which equation below represents the general term, term n , for the sequence after n hour(s), where $n \geq 0$, $n \in \mathbb{R}$?
Tick (✓) **one** box only. Give a reason for your answer.

$$\begin{array}{c} n+3 \\ \square \end{array}$$

$$3n$$

$$3^n$$

Reason:

Reason: _____

- (c)** Write down the number of bacteria you would expect to be present after 10 hours.



Question 10

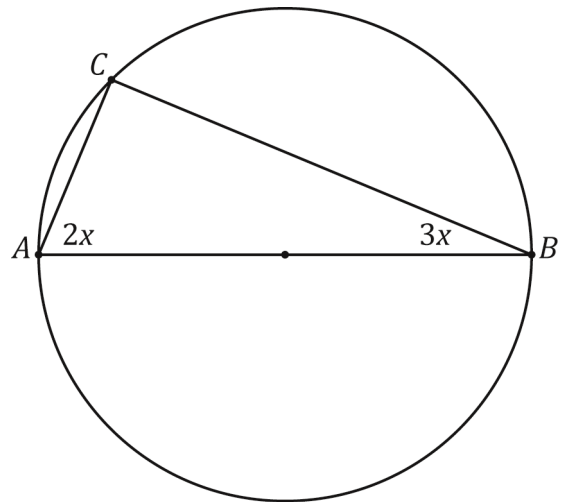
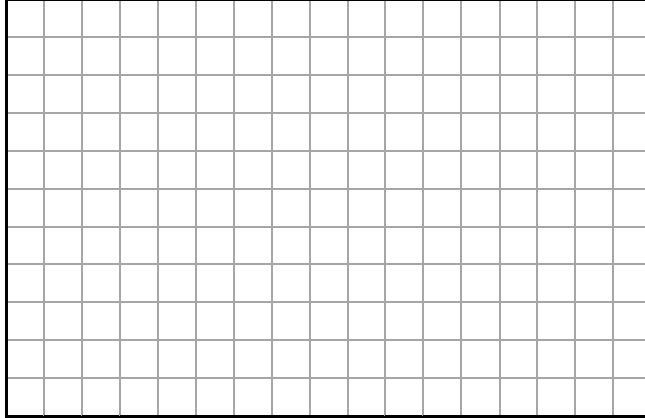
(Suggested maximum time: 10 minutes)

- (a) The diagram below shows a circle with diameter $[AB]$ (not to scale).

The point C lies on the circle.

$|\angle CAB| = 2x^\circ$ and $|\angle ABC| = 3x^\circ$.

Work out the value of x .



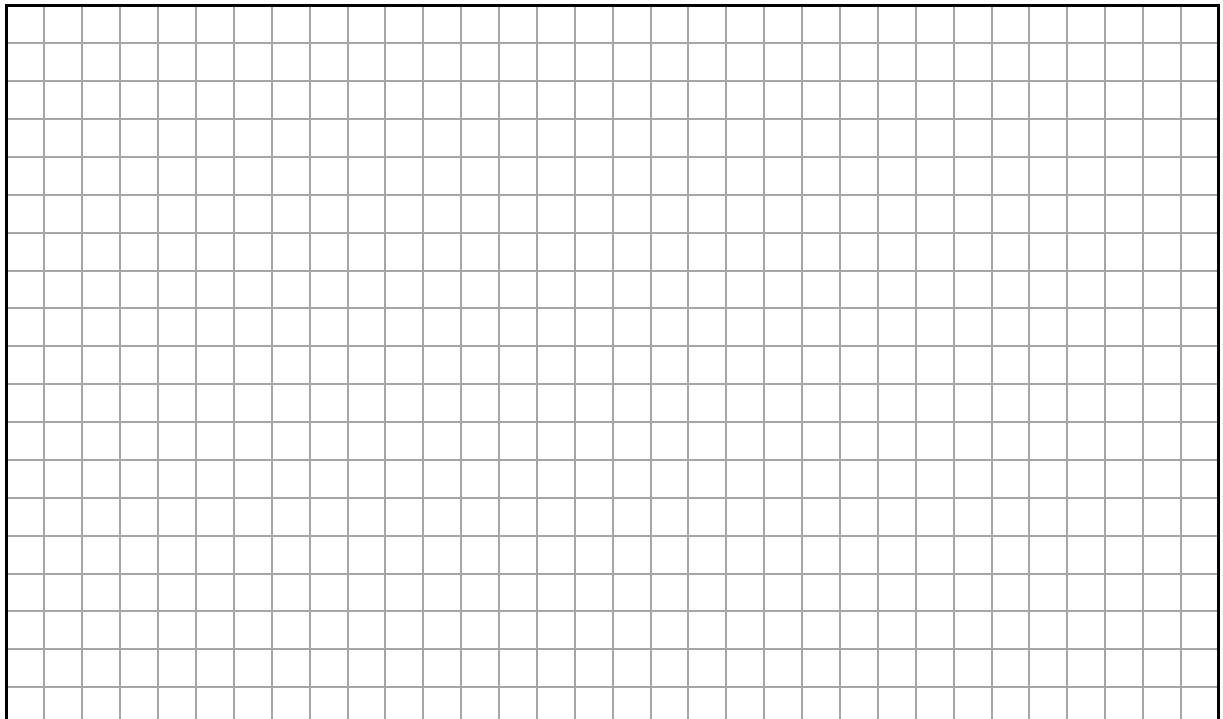
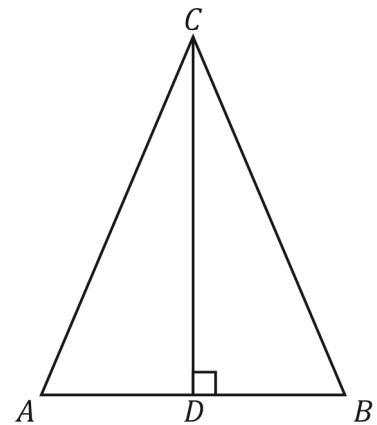
- (b) The diagram shows a triangle ABC .

The line CD is the perpendicular bisector of $[AB]$.

Prove that ABC is an isosceles triangle.

Give a reason for each statement that you make in your proof.

Hint: It may help to show that the triangles ADC and CDB are congruent.



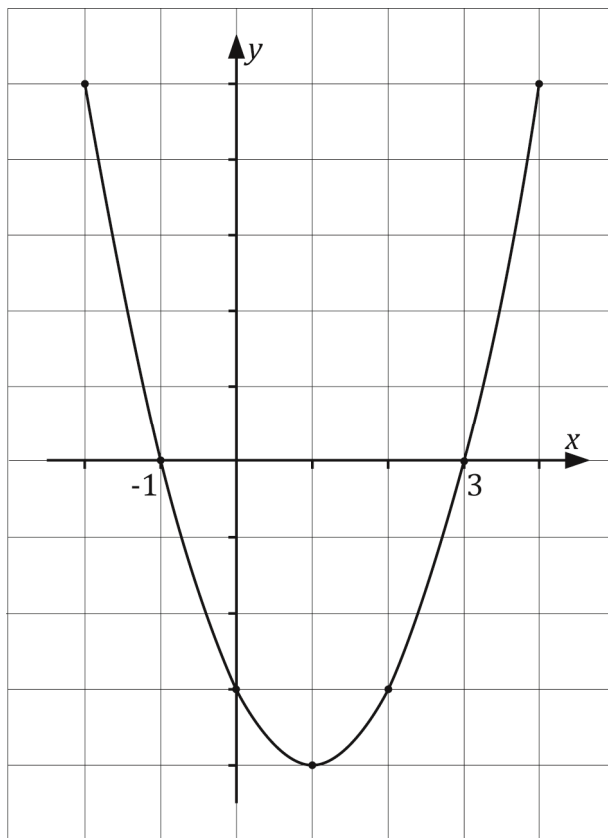
Question 11**(Suggested maximum time: 10 minutes)**

- (a) The co-ordinate diagram below shows the graph of the function

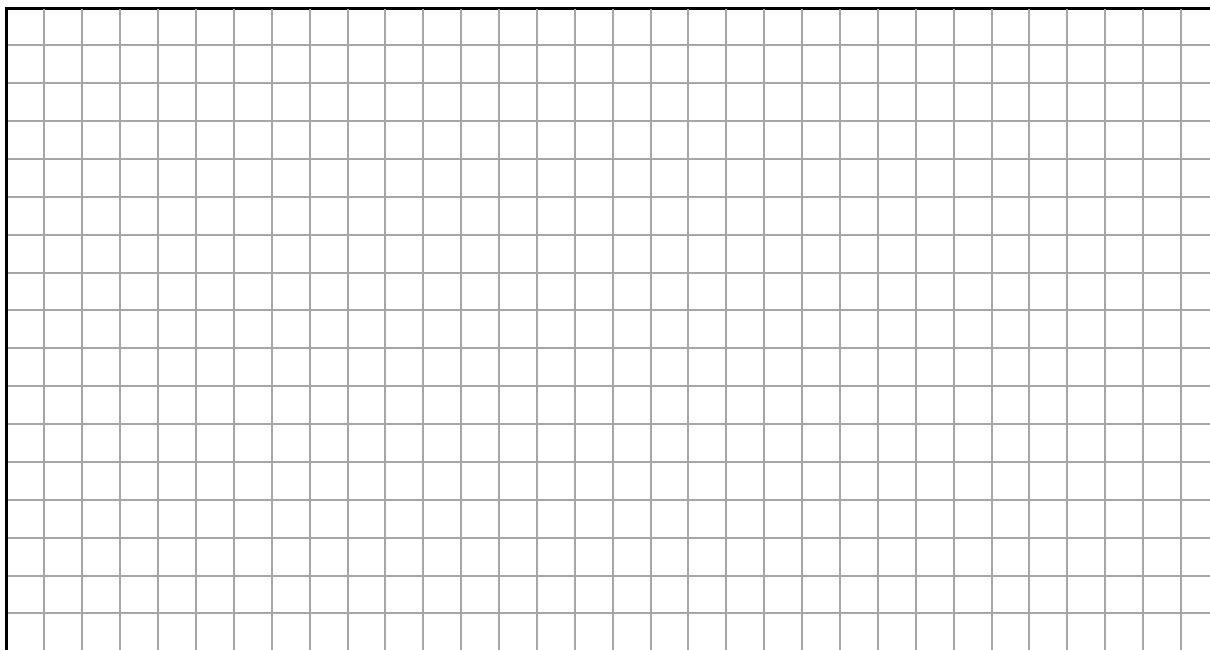
$$y = x^2 + bx + c,$$

where b and $c \in \mathbb{Z}$, for $-2 \leq x \leq 4$.

The function intersects the x -axis at $x = -1$ and $x = 3$.

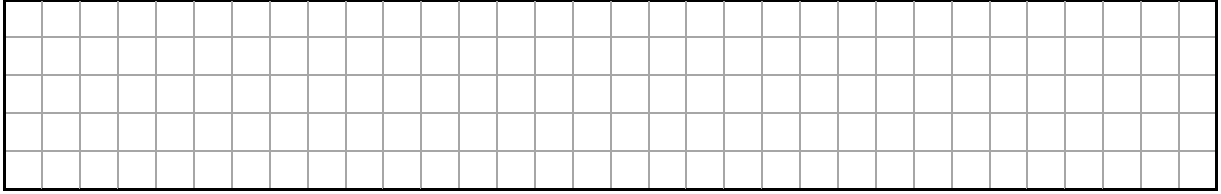


- (i) Work out the value of b and the value of c .

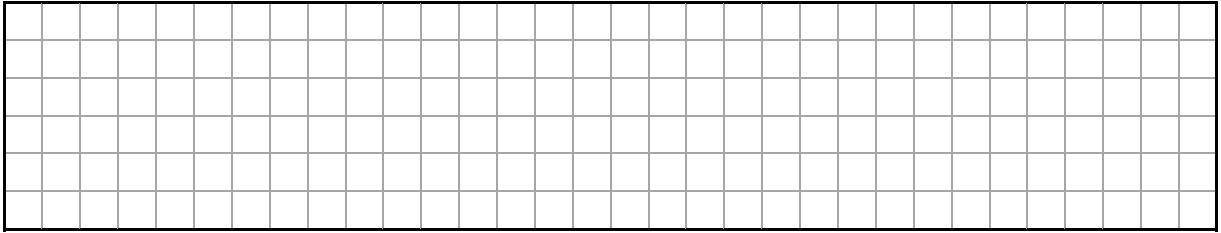


(ii) Draw the axis of symmetry of the function $y = x^2 + bx + c$ on the graph.

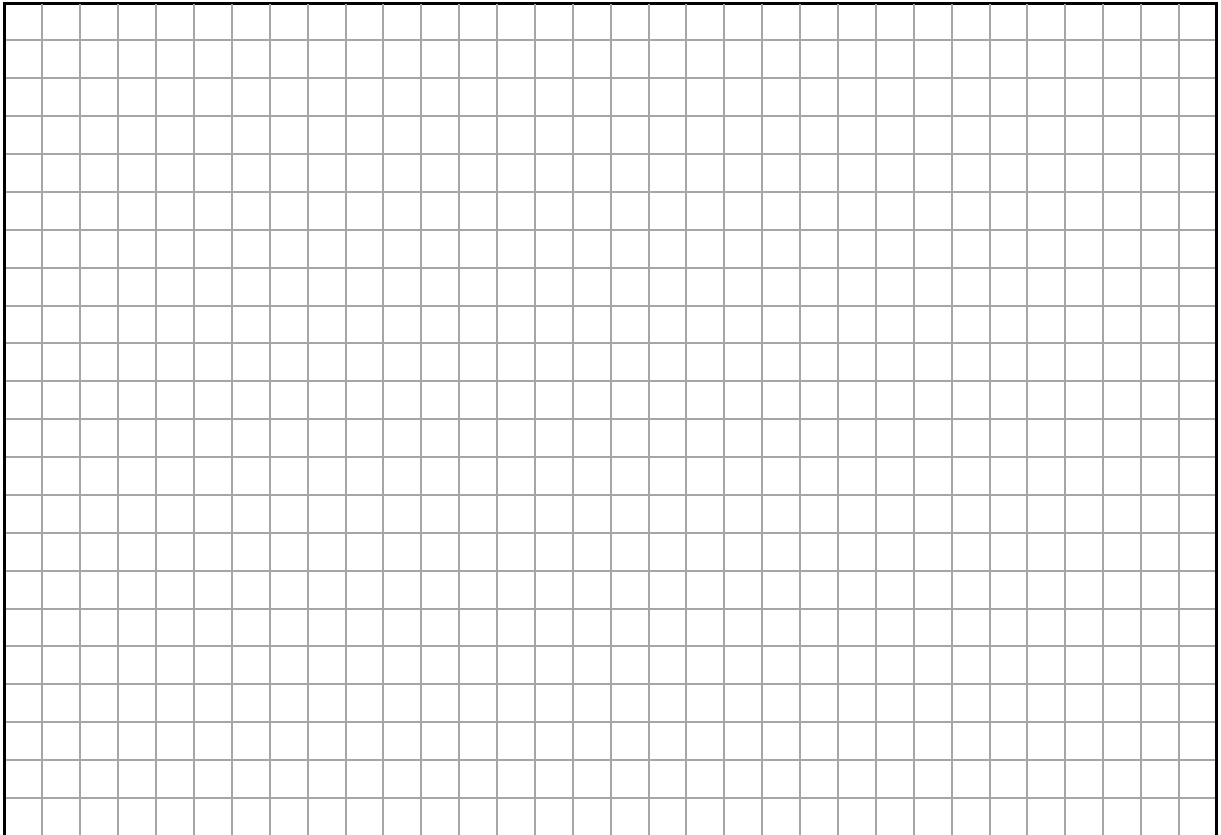
(iii) Write down the equation of this axis of symmetry.



(iv) Write down the co-ordinates of the point where the function $y = x^2 + bx + c$ intersects the y -axis.



(b) Solve the equation $3x^2 + 4x - 5 = 0$.
Give each answer correct to two decimal places.



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Pre-Junior Cycle Final Examination, 2024

Mathematics – Higher Level

Time: 2 hours

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